

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Summer Examination – 2023 Course: B. Tech. Subject Code & Name: BTEEC403 Max Marks: 60		Branch :Electrical Semester :IV Subject: Electrical Machine-II Date:18/07/2023 Duration: 3 Hr.	
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/C O)	Marks	
Q. 1	Solve Any Two of the following.	CO1	12	
A)	Explain the principal operation of 3-phase induction Motor.	CO1	6	
B)	With neat sketches discuss the construction of synchronous Motor.	CO1	6	
C)	Discuss a brief note on Construction of 3- phase induction Motor.		6	
Q.2	Solve Any Two of the following.		12	
A)	Define the following terms 1. Conductor 2. Coil 3. Turns	CO2	6	
B)	Explain Full pitch winding and short pitch winding with the help of Diagram.	CO2	6	
C)	What is pitch factor or a coil span factor (K_p or K_c)	CO2	6	
Q. 3	Solve Any Two of the following.		12	
A)	State the necessity of parallel operation of an alternator. List the condition For parallel operation.	CO3	6	
B)	Draw the Equivalent circuit and phasor diagram of a synchronous generator.	CO3	6	
C)	A 3-phase, 1500 KVA star connected 50 hz 2300 V alternator has a resistance per phase of 0.12 Ω. A field current of 70 A produces a short circuit current equal to full load current of 376 A in each line . The same field current produces an emf of 700 V on open circuit. Determine the Synchronous reactance and its full load regulation at a 0.8 P.F lagging.	CO3	6	
Q.4	Solve Any Two of the following.		12	

A)	Explain why 3-phase Induction motor never runs at synchronous Speed.	CO4	6
B)	Explain in brief production of rotating magnetic field in 3-phase induction motor.	CO4	6
C)	A 746 kw, 3-phase, 50 hz, 6 pole induction motor has a rotor impedance of $(0.02+j 0.15) \Omega$ at standstill . Full load torque is obtained at 360rpm. Calculate 1. The ratio of maximum to full load torque. 2. The speed at maximum torque. 3. Rotor resistance to be added to get maximum starting torque.	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the phenomenon of Double field revolving theory.	CO5	6
B)	Draw the circuit diagram of shaded pole motor and explain its working.	CO5	6
C)	Demonstrate the principle of operation and application of Permanent magnet motor.	CO5	6
*** End ***			